

Ada Boost

Geometric Intuition:

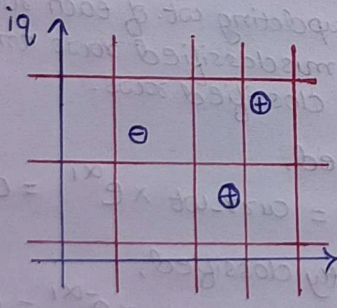
⇒ What is a weak learner? It's a classifier that performs only slightly better than random guessing.

e.g., For any yes/no problem, flipping a coin gives you 50% accuracy. A weak learner is any model that gets you 51% to 60% accuracy.

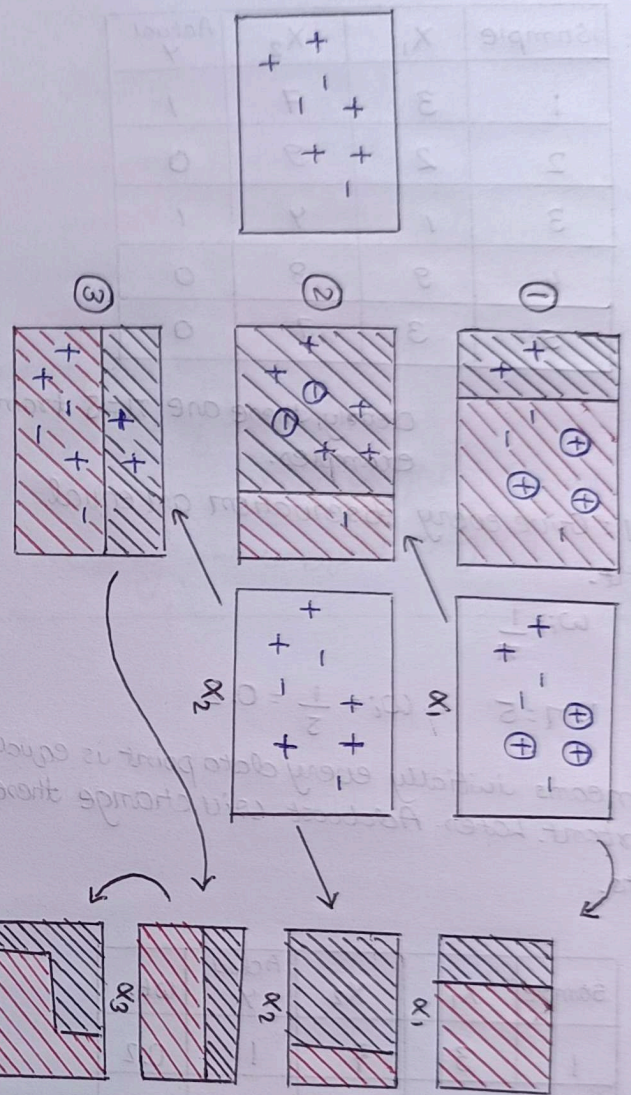
Also, they are mathematically simple, incredibly fast to train, and less likely to overfit the data on their own. AdaBoost's entire strategy is to combine hundreds of these weak learners into one strong learner.

⇒ What is a Decision Stump? It's a specific type of weak learner that AdaBoost uses by default. It's the simplest possible version of decision tree.

e.g., in decision tree we can have multiple splits but in this we make only splits in either X or Y direction and select the one which reduces most of the entropy.



⇒ What is AdaBoost? AdaBoost (Adaptive Boosting) is the most common algo that hooks these decision stumps together sequentially to build a highly accurate prediction model.



$$h(x) = \text{sign}(\alpha_1 * h_1(x) + \alpha_2 * h_2(x) + \alpha_3 * h_3(x))$$

For a new point, say {7.5, 81}

$$\text{if } \alpha_1 = 2 \rightarrow -1$$

$$\alpha_2 = 10 \rightarrow 1$$

$$\alpha_3 = 1 \rightarrow -1$$

$$\therefore h(x) = \text{sign}(2(-1) + 10(1) + 1(-1))$$

$$= \text{sign}(+ve)$$

$$= +1$$

AdaBoost Step by Step :

We have dataset as,

Sample	X_1	X_2	Actual y
1	3	7	1
2	2	9	0
3	1	4	1
4	9	8	0
5	3	7	0

Clearly, there are $n=5$ training examples.

Step 1: Give every observation an equal weight.

$$w_i = \frac{1}{n}$$

$$\because n=5 \therefore w_i = \frac{1}{5} = 0.2$$

This means initially every data point is equally important. Later AdaBoost will change these weights.

Sample	X_1	X_2	Actual y	Wt.
1	3	7	1	0.2
2	2	9	0	0.2
3	1	4	1	0.2
4	9	8	0	0.2
5	3	7	0	0.2

Step 2: Train the first weak learner. This model will give predictions, say y_{pred} . And find α and error.

Sample	X_1	X_2	Actual y	y_{pred}	Wt.
1	3	7	1	1	0.2
2	2	9	0	1	0.2
3	1	4	1	0	0.2
4	9	8	0	0	0.2
5	3	7	0	0	0.2

Clearly, the model (m_1) makes mistakes in ^{two} row. This error is measured in terms of error rate (α_1). Since, more the mistakes then the error rate should be.

$$\alpha = \frac{1}{2} \ln \left[\frac{1 - \text{error}}{\text{error}} \right]$$

$$\therefore \text{error} = \text{sum of wt. of misclassified rows}$$

$$= 0.2 + 0.2$$

$$= 0.4$$

$$\therefore \alpha_1 = \frac{1}{2} \ln \left(\frac{1 - 0.4}{0.4} \right) = 0.20$$

Step 3: We are updating wt. of each row, we will inc. the wt. of misclassified rows and dec. the wt. of correctly classified rows.

$\therefore$ for misclassified,

$$\text{new_wt} = \text{curr_wt} \times e^{\alpha_1} = 0.24$$

and for correctly classified,

$$\text{new_wt} = \text{curr_wt} \times e^{-\alpha_1} = 0.16$$

Sample	X_1	X_2	Actual y	y_{pred}	Wt.	Updated Wt.
1	3	7	1	1	0.2	0.16
2	2	9	0	1	0.2	0.24
3	1	4	1	0	0.2	0.24
4	9	8	0	0	0.2	0.16
5	3	7	0	0	0.2	0.16

$$\downarrow$$

$$\sum \text{wt.} = 1$$

$$\downarrow$$

$$\sum \text{wt.} > 1$$

updated

Step 4: We will now normalise the updated wt as their sum is greater than 1.

$$\therefore \text{Normalised wt.} = \frac{\text{New wt.}}{\sum \text{New wt.}}$$

X_1	X_2	Actual Y	Pred. Y	wt.	updated wt.	Normalised
3	7	1	1	0.2	0.16	0.166
2	9	0	1	0.2	0.24	0.25
1	4	1	0	0.2	0.24	0.25
9	8	0	0	0.2	0.16	0.166
3	7	0	0	0.2	0.16	0.166

Step 5: We will write data in form of ranges and will go for upsampling.

X_1	X_2	Y	New wt	Range
3	7	1	0.166	0-0.166
2	9	0	0.25	0.166-0.416
1	4	1	0.25	0.416-0.666
9	8	0	0.166	0.666-0.832
3	7	0	0.166	0.832-1.0

Now, we will select 5 random numbers between 0 and 1 and we will assign them in the rows in which they fall as per range.

$\therefore$ No. b/w 0 and 1:

0.13 $\rightarrow$ Row 1

0.43

0.62

0.50

0.8 $\rightarrow$ Row 4

$\rightarrow$ Row 3

$\therefore \{1, 3, 3, 3, 4\}$

$\therefore 3$ is more frequent

$\therefore$ wt. of 3 is increased

New data set will be created and new loop will be run again, when the no. of decision stump is met.

Bagging Vs Boosting:

① Types of model used:

• Bagging $\rightarrow$ LBHV

• Boosting $\rightarrow$ HBLV

② Sequential Vs Parallel:

• Bagging $\rightarrow$ Parallel. M_1, M_2, M_3 can be trained independently.

• Boosting $\rightarrow$ Sequential. M_3 depends on what M_2 did and M_2 depends on M_1 .

③ Weightage of base learners:

• Bagging $\rightarrow$ Equal weight

• Boosting $\rightarrow$ Unequal weights